

Scope for June Examination scheduled for 19th July 2021

The Scope for the Examination is from Chapter 1 to Chapter 19. The details follow below.

Scope Statement for Normal-, Supplementary-, Sick-, Aegrotat-Examination

- Chapter 1 to Chapter 19 are included in the scope.
- Everything we have covered in the live BB Collaborate sessions, whose recordings are on our ClickUP page.
- Pay particular attention to the instructions that accompany Part B of the examination. It is very similar to that in Part B of Second Semester Test.
- DEFINITIONS, REASONS/JUSTIFICATIONS, DETAILS.

- **Exclusions:**

No Section 6.3 (Motion in accelerated frames). Read and understand Section 6.3 up to before Example 6.3

No need to know Model 1 (know just the results! i.e. Equation 6.2) in Section 6.4 (Motion in the presence of a resistive force). Reading up to Equation 6.5 deepens understanding!

No System of particles

No Motion of the centre of mass

No Section 9.8 Deformable Systems

No Section 9.9 Rocket Propulsion

No Rolling motion of a rigid body (Section 10.9)

Everything after example 11.3 (page 290) up to the end of the chapter is excluded

Chapter 12 is excluded in its entirety

Chapter 13 is not excluded as such. The content is handled sufficiently in Chapter 7 already.

Chapter 14 is excluded in its entirety

No Torsional Pendulum (Chapter 15) and No Sections 15.6 Damped and

No 15.7 Forced oscillations

Section 16.3, the derivation and problems that were dealt with in class are included

No 16.4 Rate of energy transfer by sinusoidal waves in strings

No 16.5 The linear wave equation

No 16.7 Speed of sound waves

No 17.6 Standing Waves in Air Columns. **However**, students still need to be able to identify the modes of vibration similar to what we did for the string tied at both ends

No 17.7 Beats: Interference in time

No 17.8 Nonsinusoidal waveforms

No First Law of thermodynamics (ie. We end Chapter 19 with and including section 19.3).

- **Definitions**

Definitions of all kinematic parameters in 1-D and 2-D: position, displacement, velocity, and acceleration. Both instantaneous and average parameters. Analysing kinematic graphs.

Definitions in all subsequent chapters as done in the Course Notes PDF notes.

Derivations and Theory Required.

The sections below are highlights of the derivations and theory that is important. While care was taken to try and include everything, some sections may have been inadvertently left out. The class notes (or recorded lectures) are still the best reference in terms of what was done. This list is issued with that rider!

Deriving Kinematic Equations using non-calculus methods.

Deriving Kinematic Equations using calculus methods.

Deriving Dot and Cross product results in component form.

Transforming Kinematic Equations into vector forms.

Deriving the expressions for maximum height and range of a projectile

Acceleration of a particle in uniform circular motion, $a_c = v^2/r$ which is Eqn 4.21.

Galilean transformation equation, Eqn 4.29 - 4.30, including the acceleration in the two frames of reference.

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Example 5.8 weighting a fish in an elevator. These and others are not derivations in the strict sense, but the results are very important conceptually.

The Atwood machine in Example 5.9 and the Example 5.10.

Example 5.11 Experimental determination of μ_s and μ_k ; and Example 5.13

In the same vein, the following examples are considered important derivations:

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Example 6.1 The Conical Pendulum

Example 6.3 Maximum speed of a car on a curved road surface

Example 6.4 The banked roadway

Example 6.5 The Ferris Wheel

Example 6.6 Non-uniform circular motion in the vertical plane

(No Section 6.3 (Motion in accelerated frames. However, read the explanation up to just before Example 6.7)

Motion in the presence of resistive forces: Know how to set the conditions for terminal velocity, equation 6.10.

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Work Done by a spring and on a spring p159 – 160, Eqn 7.12 and 7.13

Deriving the Work Kinetic Energy Theorem p161, section 7.5 up to Eqn 7.17. There is a second method that is also given in the notes. It uses Newtons II to establish the same result as Equation 7.17.

Equation 7.18 on page 165 under Section 7.6 Potential Energy of a System

Note the results in Section 7.8 Relationship between Conservative Forces and Potential

Energy: Equations 7.25 – 7.29

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Note Eqn 8.1 and 8.2 and how to reduce them to equations appropriate for the system you are dealing with. (This is not a derivation strictly, but it is important conceptually!

Section 8.3 Situations involving Kinetic Frictional Force. The derivation on page 192 starting with Equation 8.11 and ending in Equations 8.13 to 8.15.

Note Equation 8.16 where a nonconservative force is acting on an isolated system. These are not derivations but the results are very important because they are often the starting position in solutions.

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Showing that momentum conservation for two particles follows from Newton II & III p. 213. Note the definition of Impact in Equation 9.9 and deriving an expression for Impact involving a time-averaged force in Equation 9.9 – 9.11.

Inelastic and Elastic collisions 1D in Ch 9, page 219 – 222 Equation 9.14 – 9.24. (You may be asked to derive Equation 9.21 and 9.22. Note how it is done in the class notes. Be familiar with the form of these equations too and be able to determine what happens when Equation 9.21 and 9.22 are evaluated in extreme cases).

Collisions in 2D, page 227 Eqn 9.25 – 9.27. Note Pitfall Prevention 9.4

Study Example 9.9.

Centre of mass, section 9.6 & Systems of particles, section 9.7.

Study the accompanying examples. Discrete cases and 1-D integration. The objects may be up to 3D but the integration will be limited to 1D: algebraic summation, simple 1-D integrals. Calculating moments of inertia (see Table 10.2) and Section 10.6.

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You should be able to use the Parallel-axis Theorem! Equation 10.22

Rotational Kinetic Energy, section 10.7 11.

Energy Considerations in Rotational Motion (Look at the equivalence with translational motion. Look at Table 10.3).

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Section 11.1 The vector product and torque on a particle around an arbitrary axis.

Section 11.2 (Read through, but know the definition of L). Note how equation 11.12 was derived!

Do Example 11.3 Angular momentum of a particle in uniform circular motion.

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Limit chapter 13 to the tutorial questions (or similar) that were part of the tutorial set for chapter 7.

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Motion of an object attached to a spring (section 15.1).

Analysis Model: Particle in simple harmonic motion (section 15.2)

Equations 15.1 – 15.18.

Energy of the simple harmonic oscillator (Section 15.3).

Sections 15.4 and 15.5

Simple and Physical Pendulums

Tut on Ch15 may be sent out

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Propagation of a disturbance, Section 16.1.

Analysis Model: Travelling wave, Section 16.2

Equations 16.3 – 16.13

Study the derivation of Equations 16.18. Pay particular attention to the assumptions made.

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Section 17.1 – 17.2 (including interference and standing waves)

Reflection and Transmission, Section 17.3.

Superposition and standing waves (s17.2), chapter 17.

Waves under boundary conditions. Study section 17.4

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Temperature and the Zeroth law of thermodynamics, section 18.1

The Celsius, Fahrenheit and absolute temperature scales, page 485, up to and including

Example 18.1 – 18.4.

Thermal Expansion of Solids Equation 18.4 (length, area, volume).

Macroscopic description of an Ideal Gas, Section 18.5

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Heat and Internal Energy, section 19.1 (including units of heat and the mechanical equivalent of heat)

Specific heat and calorimetry, section 19.2 (equations 19.2 -19.6)

Latent Heat, section 19.3 (including equation 19.7 - 8). If you adopt the definition from the textbook that $L=Q/\Delta m$, remember to take the convention explained under “Section 19.3

Latent Heat” page 509 that talks about which of the elements in $\Delta m = (m_f - m_i)$ is the *higher-phase material*, so that the sign to L is correct! Otherwise, make L positive if you put energy into the system and negative if the energy is taken away from your system. The latter convention should correspond with that used for Q !

~~Derivation of Equation 19.10 Work done on a gas and generalise this.~~

~~First Law of Thermodynamics from Equation 8.2~~

~~Work done in an isothermal process, Equation 19.12~~

~~Work done in an adiabatic process, Equation 19.15~~

~~Define/Describe: Isothermal, Isovolumetric, Isobaric, Adiabatic and Free expansion). Use class notes as a guide.~~

=====ENDS=====